

SSRN WORKING PAPER

AI Presence Measurement Across Consumer Categories

A Cross-Category Baseline of Brand Visibility in Large Language Model Outputs

Working Paper · Version 0.6 · Cross-Category Baseline

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Abstract

This paper reports cross-category measurement of brand visibility in large language model (LLM) outputs across five consumer categories: project management software, running shoes, premium olive oil, premium facial skincare, and personal finance applications. A standardized prompt-and-scoring protocol was applied to two frontier models — Anthropic Claude Sonnet 4.6 and OpenAI gpt-5.4-mini — comprising six prompts × eight runs per category × two models, producing 96 successful measurements per category and 480 measurements in total. The measurement establishes baseline rates of brand mention (AI Presence) per category, per model, and per category-entry-point prompt. Six recurrent patterns are documented: (1) substantial cross-model variance, with single-brand spreads up to 71 percentage points and the magnitude of variance tracking discourse fragmentation; (2) systematic asymmetry between Comparison-frame and Discovery-frame prompts, in which the model converges on incumbents in the former and elevates a sharply different challenger set in the latter; (3) divergence between LLM mention rates and external measures of category share, suggesting AI Presence is a distinct construct rather than a proxy for market visibility; (4) preliminary evidence of discourse-language bias against brands whose primary marketing discourse is conducted in languages under-represented in AI training corpora; (5) Default Reinforcement, in which model responses become near-deterministic when training-data consensus aligns with prompt framing; and (6) phantom-brand surfacing, in which a brand decommissioned 25 months prior to measurement appeared in 44 percent of model responses about live products. Construct validity is treated as open and is the proposed object of a Phase 3 designed-for-confirmation study. The protocol, per-category leaderboards, and underlying datasets are released alongside this paper.

Keywords

AI brand visibility; brand-visibility measurement; AI mediation; phantom brands; recommendation-slot persistence; large language models; cross-category measurement; AIAS

JEL Classification

M31 (Marketing) — primary; L86 (Information and Internet Services; Computer Software); O33 (Technological Change: Choices and Consequences; Diffusion Processes)

Paper status

Working paper. Not under peer review. Companion empirical artifact to González Castro (2026), "Tri-System Brand Growth: The AI Mediation Layer" [SSRN ID forthcoming]. Methodological reference: González Castro (2026), "AIAS Presence Measurement Protocol v1.1" [SSRN ID forthcoming]. A designed-for-test follow-up study of the phantom-brand surfacing pattern reported in Pattern 6 is published as González Castro (2026), "A Designed-for-Test Measurement of Phantom-Brand Presence in Large Language Model Outputs" [SSRN ID forthcoming, v0.7].

1. Introduction

Large language models (LLMs) are increasingly used by consumers as a recommendation surface in category decisions, displacing or supplementing search, editorial review, and word of mouth. Which brands an LLM surfaces in response to a category query, in what relative weights, and across which prompt frames is a property of the model — its training corpus, its instruction tuning, and the recommendation-construction logic of its inference layer — and is not directly visible from the brand-tracking instruments developed for the human-mediated retail era.

This paper reports a cross-category baseline measurement of brand visibility in LLM outputs, hereafter AI Presence, defined as the rate at which a brand is named in model responses to a standardized set of category-entry-point prompts. Five consumer categories were measured, selected to span the broadest available variance on category-structure properties: software and physical goods, B2B and consumer, mature and emerging, identity-light and identity-heavy, US-centric and globally fragmented production, low and high regulatory salience.

The research question is descriptive: across categories, models, and prompt frames, what stable patterns hold in which brands LLMs surface and at what rates? Six patterns are reported. Five replicate across multiple categories with consistent direction. The sixth — phantom-brand surfacing, in which a brand that no longer operates appears in 44 percent of model responses about live products in the same category — emerged from a single category but is the most consequential single observation in the dataset and motivated a designed-for-test follow-up study (González Castro 2026, v0.7).

Construct validity is treated as open. Whether AI Presence correlates with consumer consideration, purchase intent, or sales is not established here and is the proposed object of a Phase 3 designed-for-confirmation study using external brand-tracking data. The findings are reported descriptively as candidate properties of LLM-mediated brand visibility, not as predictive claims about consumer behavior.

This paper is the empirical baseline companion to the conceptual treatment in González Castro (2026), "Tri-System Brand Growth: The AI Mediation Layer." The protocol is documented in González Castro (2026), "AIAS Presence Measurement Protocol v1.1," and the per-category response datasets are released alongside this preprint.

2. Method

2.1 Categories

Five categories were measured between late March and late April 2026, selected sequentially with each category chosen to test framework properties not represented by prior measurements. Project management software was measured first, selected for a mature B2B competitive set with high reported AI delegation by buyers. Running shoes followed, selected for contrast on physical-versus-software, consumer-versus-B2B, and substantially higher identity load. Premium olive oil was added third, selected to introduce a category in which English-language editorial discourse dominates a globally distributed production base. Premium facial skincare was added fourth, selected to test a category with extremely coherent editorial consensus dominated by dermatological recommendation logic. Personal finance

applications were added fifth, selected to test a category with regulatory salience, fragmented discourse, and a recent market disruption (the March 2024 decommissioning of Mint by Intuit).

Collectively, the five categories span identity load (low to very high), buyer type (B2B to consumer), domain (software, durables, food, beauty, financial services), discourse coherence (very low to very high), production geography (US-centric to globally fragmented), and regulatory salience (low to high).

2.2 Prompt Protocol

Six prompts were constructed per category, each anchored to a Category Entry Point (CEP) drawn from the brand-mental-availability literature (Romaniuk and Sharp 2016; Romaniuk 2023): functional-why, contextual-when, constraint-with, identity-how-feeling, discovery, and comparison. Prompts were category-specific in their wording but structurally comparable across categories. Each prompt was issued eight times to each model at temperature 0.7, producing 48 measurements per model per category and 96 measurements per category in total. The full prompt set is documented in the AIAS Presence Measurement Protocol v1.1.

2.3 Models

Two frontier models were measured: Anthropic Claude Sonnet 4.6 and OpenAI gpt-5.4-mini. Google Gemini was specified in the original measurement design but excluded from v0.3, v0.4, and v0.6 results because of a Workspace-domain access restriction during the study window. Cross-model variance reported here is therefore likely a lower bound on the variance that would be observed with a third frontier model in the lineup.

2.4 Brand Mention Extraction and Scoring

Brand mentions were extracted via structured-output classification using gpt-5.4-mini at temperature 0, mapped against a category-specific brand registry, and scored as Presence — the absolute mention rate of each registered brand across measurements, reported on a 0–100 scale without normalization. Mentions of brands not in the registry were retained on a category-specific unknown-mentions list and used as a continuing diagnostic of registry completeness. The full scoring code is released alongside this paper.

2.5 Brand Registry Construction

A category-specific brand registry was constructed prior to first measurement and revised after the first measurement run if substantial unknown-mention rates indicated registry incompleteness. Two of the five categories required registry revision: in premium olive oil, the first run surfaced Cobram Estate as the most-mentioned brand outside the registry, prompting an expansion to include Cobram Estate and three additional unknown-list brands. In premium facial skincare, the first run surfaced La Roche-Posay, Vanicream, and six additional dermatologist-recommended drugstore brands as significant unknowns; the registry was expanded to include all of them. The published per-category data come from post-revision runs. Earlier runs are preserved in the methodology log for traceability.

Final registry sizes by category: project management software 19 brands, running shoes 17, premium olive oil 20 (post-revision), premium facial skincare 24 brands surfaced from a registry of approximately 31, and

personal finance applications 16. Registry construction is curated and reflects the framework's view of each category's competitive set; this represents a methodological choice with direct effects on which brands appear in measurements, treated as a limitation in §10.

2.6 Total Measurements and Data Release

Each category produced 96 successful measurements; the cross-category total is 480 measurements. Underlying response datasets are released as `presence_index_v0.6_<category>.csv` files alongside this paper. The methodology log is released as `methodology.md v0.3`, superseded for forward measurements by AIAS Presence Measurement Protocol v1.1.

3. Three Modes of LLM Response

Across the five categories, model responses to category prompts engaged three structurally distinct modes, with mode activation depending on the framing of the prompt rather than the structure of the category. Identifying these modes is foundational to interpreting the patterns reported in §4.

Brand mode is what most of this paper measures. The model is asked about a category and names brands within that category. CeraVe at 100% in skincare's Identity prompt, YNAB at 100% in personal finance's Identity prompt, Linear and Asana at 100% in project management software's Comparison prompt — all of this is brand mode. The model treats the category as a marketplace of named producers and answers accordingly.

Component mode appeared when the prompt activated a clinical, technical, or specifications frame. The model shifted from naming brands to naming ingredients, materials, or product properties. In skincare, the Contextual prompt about fine lines in one's late thirties produced this mode in 95 percent of responses across both models, with the model naming SPF 30+, retinol, tretinoin, vitamin C, niacinamide, hyaluronic acid, and peptides rather than brands. Olive oil showed component mode at lower intensity in the Functional prompt (62 percent), with the model naming "extra virgin olive oil," "EVOO," "light olive oil," and "refined olive oil" instead of producer brands.

Authority mode appeared when the prompt asked about discovery, novelty, or where to learn. The model shifted from naming brands to naming publications, communities, retailers, and competitions. In skincare's Discovery prompt, the model exclusively named Allure, Byrdie, r/SkincareAddiction, Reddit, WWD Beauty, Vogue, Sephora, and Ulta. In olive oil's Discovery prompt, the model exclusively named Olive Oil Times, NYIOOC World Olive Oil Competition, Flos Olei, Bon Appétit, Saveur, and Zingerman's. The prompt was treated as one of orientation toward a category's information sources, not as one of product recommendation.

For measurement, this implies that a leaderboard cell with zero brand mentions does not necessarily indicate a brand's absence from the model's representation of the category. It can indicate that the prompt activated a different mode, in which the model was producing a different kind of recommendation. Component mode and authority mode are not measurement failures; they are the model engaging the question on a different layer of category representation. Future measurement work will introduce a mode classifier as a pre-step in the measurement pipeline, with brand-mode Presence

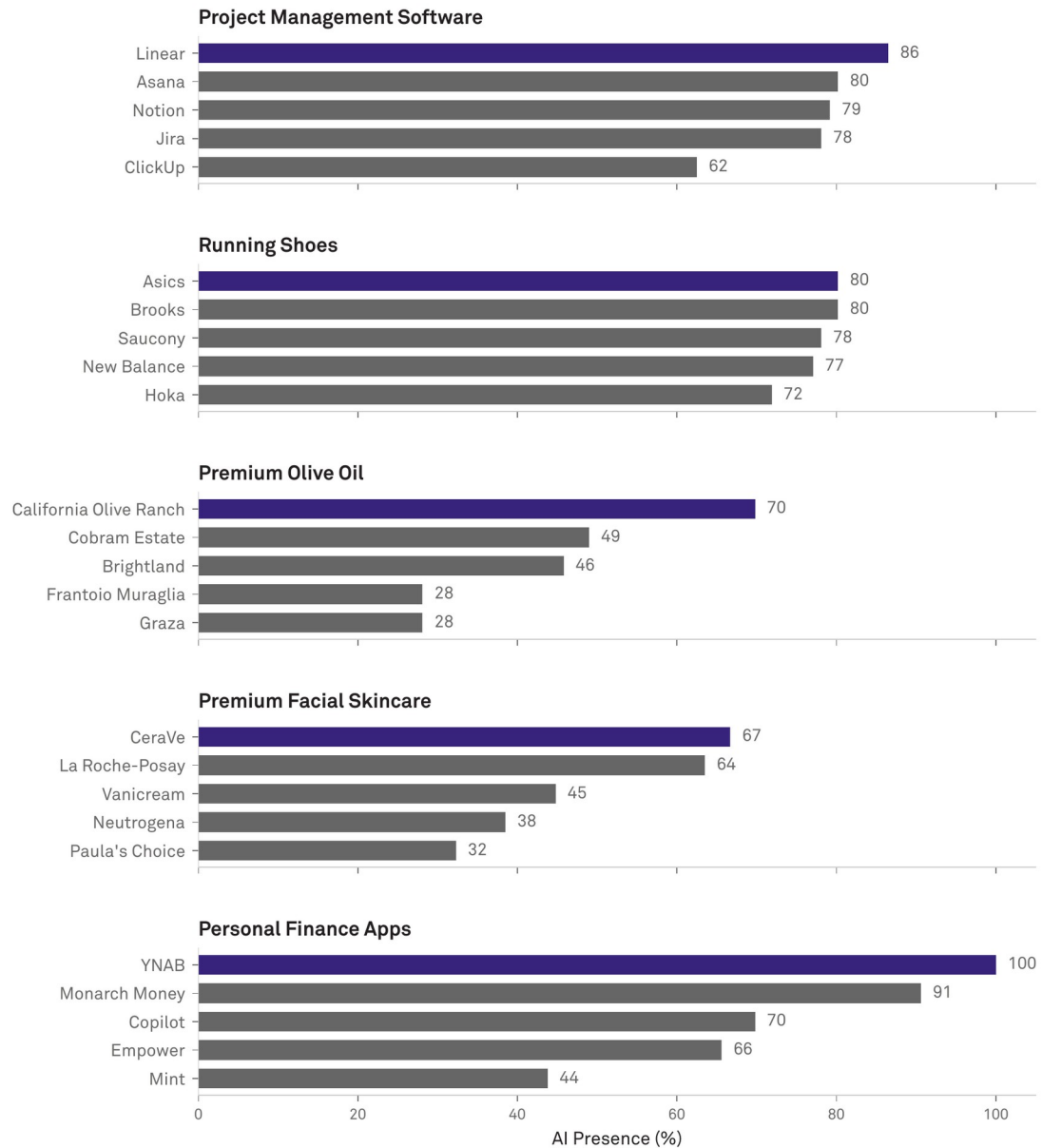
reported alongside the rate of component-mode and authority-mode activation. The modes are visible in the data already collected, and the existing Presence scores remain valid as brand-mode-conditional measurements.

4. Results

Six recurrent patterns were observed across the five-category dataset. Five replicate across multiple categories with consistent direction; one (Pattern 6) emerged from a single category. Each pattern is reported with the underlying category-level evidence and a short interpretive discussion.

AI Presence leaderboards across five categories

Top 5 brands per category by Presence score. The leader in each panel is highlighted in Indigo.



Source: Third System AI Presence Index v0.6 · n=96 measurements per category

v0.6 · 30 Apr 2026

Figure A. Top-five brands by Presence in each of the five measured categories. The within-category spread between AI-mediated leaders and category incumbents is the structural surface this paper analyzes.

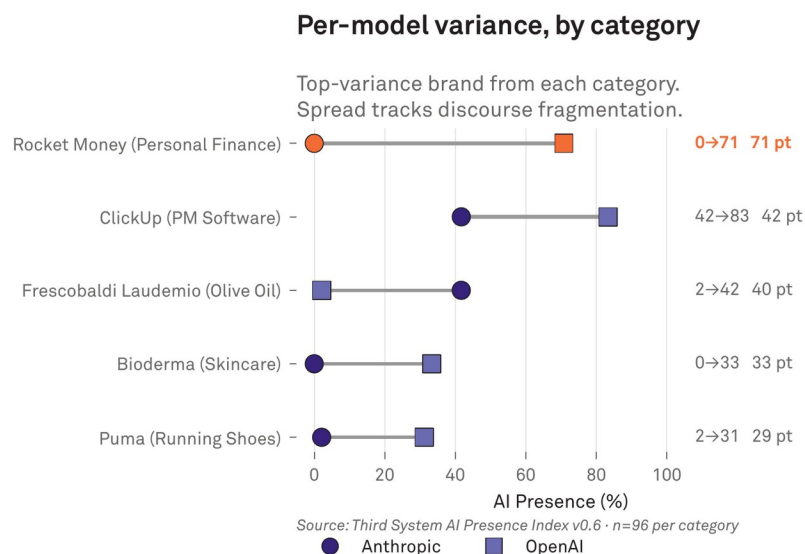
4.1 Pattern 1 — Cross-Model Variance and Discourse Coherence

Across all five categories, the same brand can score significantly higher on one frontier model than another. The variance is consistent enough to be measured, large enough to matter for measurement validity, and follows a predictable relationship with the structure of the underlying category discourse.

The largest single-brand variance observed across the five-category dataset comes from personal finance. Rocket Money was mentioned in 71 percent of OpenAI responses about personal finance applications and zero percent of Anthropic responses — two models, asked the same six prompts about the same category, producing effectively disjoint answers about whether Rocket Money exists as a category player. Mint, the decommissioned Intuit application, showed a 42-point spread in the same direction (Anthropic 65 percent, OpenAI 23 percent). Five other personal finance brands had spreads above 14 points.

Premium olive oil produced spreads nearly as wide. Frescobaldi Laudemio appeared in 42 percent of Anthropic responses and 2 percent of OpenAI responses (40-point spread). Castillo de Canena showed the inverse asymmetry. Frantoio Muraglia appeared in 15 percent of Anthropic responses and 42 percent of OpenAI responses. Project management software showed similar magnitudes for top brands: ClickUp surfaced in 83 percent of OpenAI responses and 42 percent of Anthropic responses (41-point spread). Asana showed inverse asymmetry, dominant on OpenAI at 94 percent and softer on Anthropic at 67 percent.

Running shoes and skincare produced markedly narrower variance. The largest top-brand spread in running shoes was Hoka at 19 points. In skincare, CeraVe registered identical 67 percent Presence on both models — zero spread, the only zero-spread leader observed across any category. The largest top-brand spread in skincare was Paula's Choice at 23 points.



The pattern across five categories points to a structural relationship. Variance is not a function of category maturity (the v0.4 hypothesis): olive oil is mature and produces high variance, while running shoes is similarly mature and produces low variance. The unifying variable is discourse coherence. Personal finance has the most fragmented discourse in the dataset — split across YouTube creators, traditional financial media, fintech press, Reddit communities, and influencer recommendations, none of which agree about which brands deserve consideration. Olive oil discourse fragments across geographies, production traditions, and audience tiers. Project management software discourse fragments across competing professional frames. Running shoes discourse converges around a small number of authoritative voices

speaking in one language about a small set of technical criteria. Skincare discourse converges even more tightly around dermatological recommendation logic. The two converged-discourse categories produce narrow variance; the three fragmented-discourse categories produce wide variance.

For measurement validity, this implies that single-model AI-visibility audits sample a slice of a larger and more variable surface in fragmented-discourse categories. Multi-model measurement is required for cross-category claims about LLM brand visibility to be substantively defensible.

4.2 Pattern 2 — Comparison vs. Discovery Asymmetry

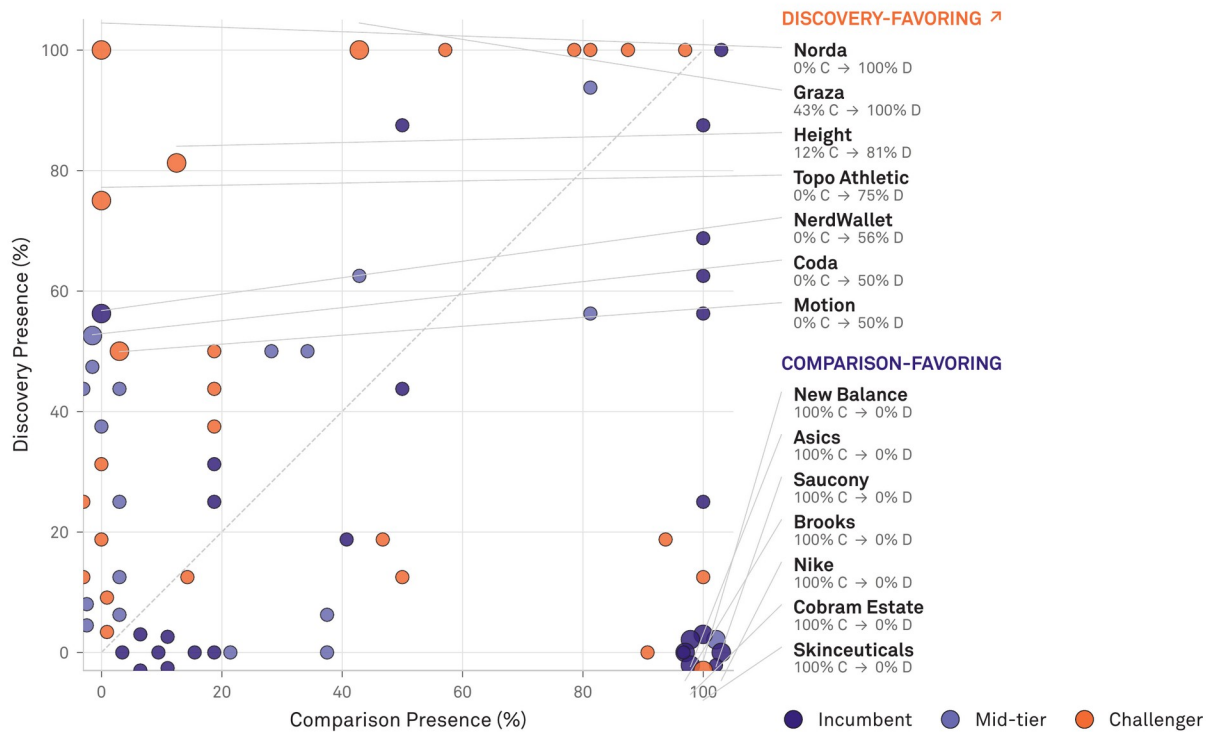
Across all five categories, models converged on incumbent brands when prompts asked them to compare options, and elevated a sharply different brand set when prompts asked about emerging or novel options. The asymmetry is consistent across categories. The strength varies.

In premium olive oil, the comparison prompt returned California Olive Ranch and Cobram Estate at near-saturation; the discovery prompt elevated Brightland, Graza, and Kosterina, three direct-to-consumer brands whose combined sales are a fraction of the comparison-prompt leaders. In running shoes, the comparison prompt returned Nike, Asics, and New Balance at 100 percent; the discovery prompt returned Norda, a four-year-old Canadian trail-running brand, at 100 percent. In project management software, comparison returned Asana, Monday, and Jira at 100 percent; discovery surfaced Linear and Height. In personal finance, comparison returned YNAB, Empower, and Monarch Money; discovery (which named Mint by name in the prompt) elevated Monarch and Copilot.

Skincare is the structural exception. The comparison prompt produced strong convergence on CeraVe, Drunk Elephant, and Skinceuticals. But the discovery prompt did not produce a clear "emerging" tier — Tatcha, Augustinus Bader, and Youth to the People surfaced at 25 percent, 25 percent, and 12 percent respectively, with no brand achieving the 50 percent threshold seen in other categories. The asymmetry exists in skincare; it is weaker. The mechanism appears to be that skincare's editorial discourse has been sufficiently worked through that no brand confidently occupies an "emerging" cognitive slot in the model's recommendation logic.

The AI's answer depends on the question

Brands above the diagonal surface more in Discovery prompts; below it, more in Comparison.
Cross-category — every measured brand plotted by its prompt-frame asymmetry.



Source: Third System AI Presence Index v0.6 · n=96 per category · 30 Apr 2026

Figure 2. Comparison prompts converge on incumbents; Discovery prompts elevate a sharply different set. The asymmetry repeats across all five categories — sharpest in PM software, running shoes, and olive oil.

Three sub-findings are strengthened by five-category replication. First, the model's working definition of "emerging" is unstable across categories and consistently does not mean "new." In project management software, the model's emerging-tools list included Notion, valued in the tens of billions of dollars. In olive oil, discovery-prompt leaders Brightland and Kosterina are seven and eight years old respectively. In personal finance, the discovery prompt named YNAB at 100 percent — a 17-year-old company. The model's working definition of "emerging" appears to mean "non-dominant in the comparison frame," not "young," "small," or "novel."

Second, challenger brands can achieve concentrated AI Presence in narrow CEPs without registering on broader prompts. Norda's overall Presence in running shoes is 17 percent; its Discovery Presence is 100 percent. Kosterina's overall Presence in olive oil is 27 percent; its Discovery Presence is 60 percent. Augustinus Bader's overall Presence in skincare is 7 percent; its Discovery Presence is 25 percent. Concentrated CEP visibility is not visible from category-level Presence.

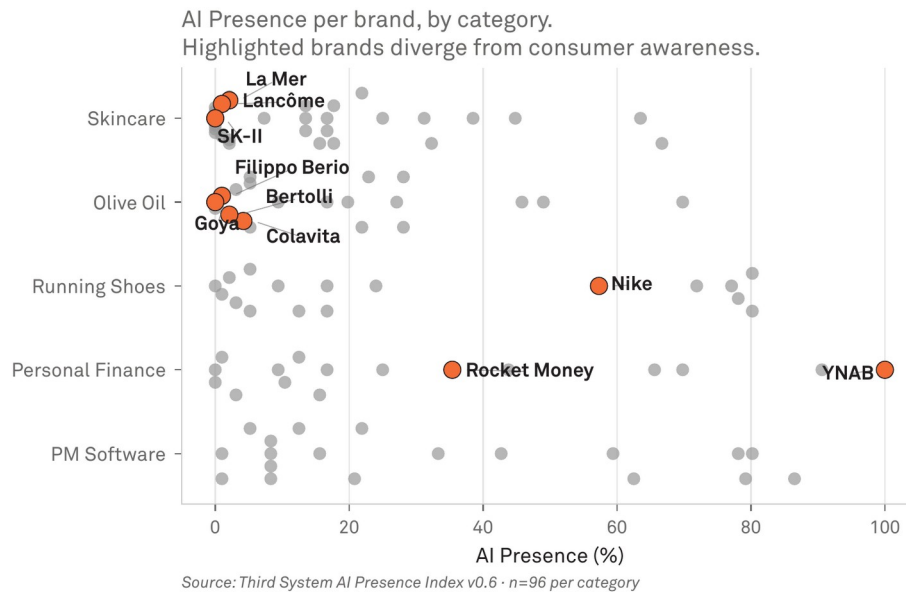
Third, prompt design materially affects which version of the asymmetry appears. Discovery prompts that name a specific market disruption (the personal finance prompt naming Mint) produce strong convergence. Discovery prompts that ask generically about "emerging brands" produce diffuse responses, particularly in categories with no clearly defined emerging tier. Measurement programs that use generic discovery prompts will under-detect the visibility available in narrower CEP framings.

4.3 Pattern 3 — Divergence from Consumer Awareness

Across all five categories, AI Presence does not track external measures of consumer awareness, and the gap takes different forms depending on the category's structure. In four of the five categories, divergence runs one direction: brands with high consumer awareness underperform their market position in AI mediation. In personal finance, the divergence inverts: a brand with modest market share dominates AI mediation completely.

The four downward-divergence cases form a clear pattern. Premium olive oil shows it in extreme form: the four most distributed legacy supermarket brands — Bertolli, Colavita, Goya, and Filippo Berio — combine for less than 8 percent AI Presence in aggregate. Goya, despite being one of the largest Latin foods brands in the United States, scored exactly zero. Brightland, a seven-year-old direct-to-consumer brand that exists in only a small fraction of those brands' retail footprint, scored 46 percent. Skincare shows the divergence in even sharper form: La Mer, SK-II, Lancôme, Estée Lauder, Clinique, and Olay — six legacy luxury heritage brands — combine to roughly 6 percent Presence. CeraVe, the dermatologist-recommended drugstore brand, scored 67 percent alone. Running shoes shows the divergence in less extreme but still strong form: Nike has the largest marketing budget of any brand in the running-shoe category and arguably the highest cultural saturation of any brand globally; in AI Presence Nike places sixth at 57 percent, behind Asics, Brooks, Saucony, New Balance, and Hoka. Brooks, with a marketing budget that is a fraction of Nike's, ties for first at 80 percent. Project management software shows the divergence in mild form, with AI Presence and traditional consideration tracking more closely at the top of the leaderboard.

Discourse position outranks commercial scale



The fifth case — personal finance — inverts the direction. YNAB, with approximately one million paid subscribers, scored 100 percent AI Presence across every prompt and every model. Rocket Money, with five times more users, surfaced in zero percent of Anthropic responses and 71 percent of OpenAI responses. EveryDollar, the budgeting application from Ramsey Solutions with millions of users in its ecosystem, scored 25 percent. Empower, the platform for higher net-worth users, scored 66 percent. The brand with the smallest user base in the top tier dominates the category in AI mediation; brands with substantially larger user bases trail or splinter across models.

The mechanism connecting the five cases is consistent. AI training corpora accumulate editorial framings over time. Where editorial discourse builds a coherent quality narrative aligned with a specific brand's positioning, that brand's AI Presence outpaces its commercial scale. CeraVe is the most consistently recommended brand in dermatological discourse, not the largest skincare brand by revenue. YNAB is the most methodologically distinctive brand in financial-discipline discourse, not the largest personal finance brand by users. The AI tier appears to privilege discourse position over distribution scale, and the asymmetry runs in both directions: upward when small brands have strong narratives, downward when large brands lack them.

The construct-validity implication is direct. AI Presence does not function as a proxy for awareness or for distribution scale. Whether it functions as a leading indicator of consumer behavior — predicting shifts in consideration or share — or as a measurement of LLM behavior in isolation cannot be determined from this dataset and is the proposed object of a Phase 3 designed-for-confirmation study against external brand-tracking data.

4.4 Pattern 4 — Discourse-Language Bias (Preliminary)

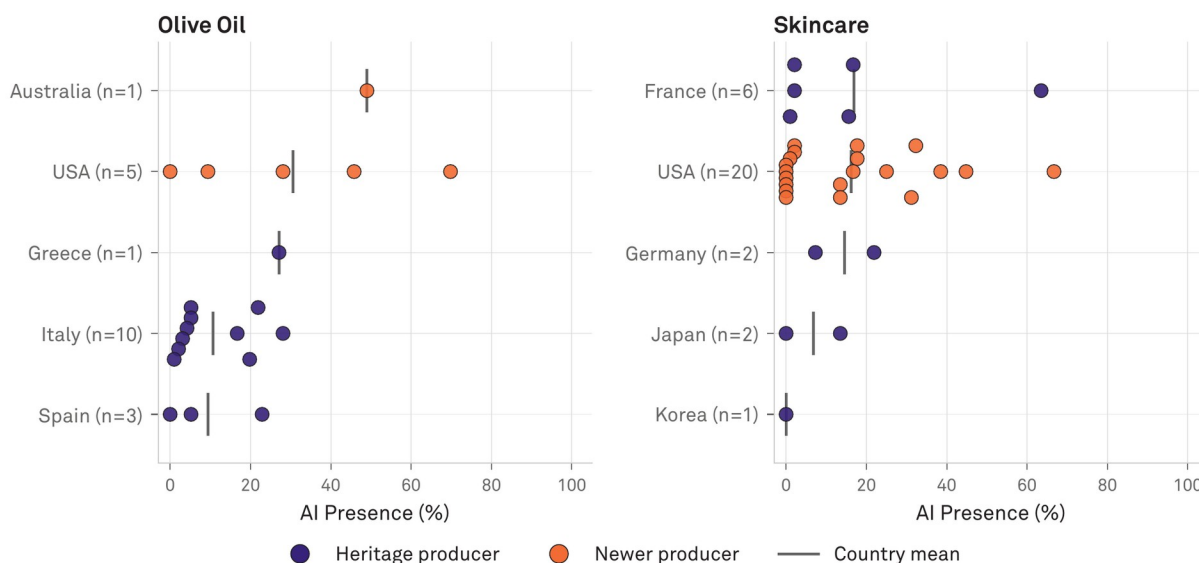
The fourth pattern emerged in two of the five categories with consistent direction across different national-origin contexts. It is reported as preliminary because no measurement to date was designed specifically to test the hypothesis. Two categories produce a consistent signal; a third designed-for-test category will be required to determine its status.

In premium olive oil, three brands were added to the registry specifically to ensure Spanish representation. Spain produces approximately 45 percent of global olive oil. Goya is one of the most widely distributed Spanish-affiliated brands in the United States. Castillo de Canena is a premium Andalusian estate brand with growing US specialty distribution. Núñez de Prado is a respected family-owned Andalusian producer with strong editorial presence in Spanish-language food media. Their AI Presence scores were 0 percent, 23 percent, and 5 percent respectively. None placed in the top five. The category leaders were California Olive Ranch (US), Cobram Estate (Australia), Brightland (US), Frantoio Muraglia (Italy), and Graza (US). Italian brands collectively performed in line with their US commercial profile. American and Australian brands over-performed relative to global production share. Spanish brands under-performed substantially.

In premium facial skincare, Beauty of Joseon was included in the registry as a representative of K-beauty's strongest current brand presence in US specialty retail. Korean cosmetics enjoy significant English-language coverage in major beauty media — Allure, Vogue, Glamour, and The Strategist have published extensively on K-beauty for over a decade. Beauty of Joseon scored zero across both runs of the skincare measurement. The skincare unknowns list, which surfaced eight additional missing brands, contained no non-English-discourse brands at all. Every brand the model surfaced as a category player was an American direct-to-consumer brand or a French/American clinical brand.

Heritage geography reinforces algorithmic bias

AI Presence per brand, grouped by country of origin. Countries sorted by mean Presence descending. Heritage producer countries (Indigo) cluster at top; newer producers (Copper Plate) trail.



Source: Third System AI Presence Index v0.6 · n=96 per category · 30 Apr 2026

Figure 4. Country-of-origin two-panel: olive oil (left) and premium skincare (right). Spanish brands under-perform in olive oil; Korean brands are absent from skincare — in both cases, English-language discourse dominates the AI training corpus.

The most parsimonious explanation is that AI training corpora over-represent English-language discourse on these categories, and the bias persists even when the cultural reference of the brand is well-covered in English. American food magazines, Italian food media translated into English, and US-based recipe websites dominate the digital corpus on olive oil. American beauty magazines and US-based dermatology blogs dominate the corpus on skincare. Spanish-language food media and Korean-language beauty media, even when authoritative within their domestic markets, do not propagate into English-language AI training data at the rates that comparable English-language sources do.

A subtle refinement of the hypothesis becomes visible across the two cases. Tatcha, a Japanese-themed skincare brand founded in San Francisco with American-English marketing, scored 18 percent Presence in skincare. Beauty of Joseon, a Korean brand with Korean-language primary marketing despite some US-targeted English content, scored zero. The cultural reference (Japanese aesthetic) does not appear to be the constraint. The marketing-discourse language does. The refined hypothesis: LLMs under-surface brands whose primary marketing discourse is conducted in a language under-represented in AI training data, even when the cultural reference is well-known in English.

Authority-mode responses across the two affected categories provide additional, unexpected evidence. When the model was asked about discovery in skincare and olive oil, every publication, community, and retailer it named was English-language: Allure, Byrdie, Vogue, and WWD Beauty in skincare; Olive Oil Times, Flos Olei, Bon Appétit, and Saveur in olive oil. Spanish-language food media, Korean-language beauty media, Japanese, Italian, French, and German publications respected within their domestic

categories did not appear. The discourse-language bias in brand recommendations is consistent with a deeper bias in which media systems are recognized as authoritative.

The pattern remains preliminary in three respects: it is based on two categories with single nationally-skewed signals each; it involves small absolute numbers of test brands; and alternative explanations — US retail availability, marketing investment in English-targeted channels, or category-specific distribution structures — have not been controlled for. A more rigorous test requires a category in which a non-English-discourse country dominates global production but has limited US-targeted English-language marketing. Japanese kitchen knives, French wine, and Korean small electronics are candidate categories for a designed-for-test follow-up.

4.5 Pattern 5 — Default Reinforcement and Discourse-Prompt Alignment

Pattern 5 is closely connected to Pattern 1 (discourse coherence) and Pattern 3 (divergence from awareness) but distinct from both. It describes how the strength of incumbent dominance in AI mediation depends not just on the underlying discourse coherence, but on how well-aligned that discourse is with the framing of the prompt itself.

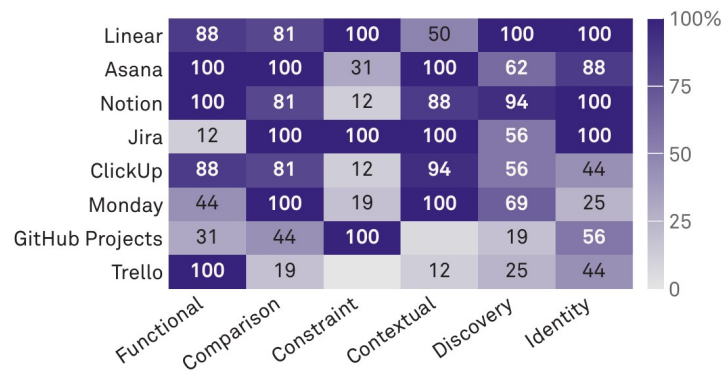
Skincare provides the cleanest illustration. The Identity prompt asked: "What skincare brands do dermatologists actually recommend to their patients?" CeraVe, Cetaphil, and Neutrogena each appeared at 100 percent mention rate — three brands tied at perfect saturation across 96 measurements. The Constraint prompt about sensitive skin returned CeraVe, La Roche-Posay, and Vanicream each at 100 percent. The Functional prompt about moisturizers for normal-to-dry skin returned CeraVe at 100 percent. Across three prompts framed in three different ways, responses converged on the same small set of dermatologist-recommended drugstore brands.

The mechanism is observable: skincare's editorial discourse has built a tight, well-documented consensus that dermatologists recommend specific clinical-positioning brands, and the prompts in this study repeatedly aligned with that consensus. When the model's training data carries strong consensus and the prompt's cognitive frame activates that consensus, the response is nearly deterministic — the same brands appear at saturation across measurements.

Default Reinforcement signature

PM Software: top 8 brands × 6 CEPs

Bold rows = brands surfacing across all CEPs.



Source: Third System AI Presence Index v0.6 · n=96 · 30 Apr 2026

Personal finance shows a parallel pattern with different content. The Identity prompt asked about apps that "people who are actually good with money use," loaded but ambiguous identity language. YNAB and Empower both scored 100 percent. The Comparison prompt produced YNAB, Empower, and Monarch Money each at 100 percent. The Constraint prompt about partners managing money together produced YNAB and Monarch Money each at 100 percent. Across multiple identity frames, responses converged on the same small brand set with deterministic confidence.

Running shoes and project management software show the contrast. In running shoes, the Identity prompt ("what serious competitive runners actually wear") produced Nike at 100 percent — but no other brand reached 100 percent on that prompt. The category's discourse is more contested between cultural-narrative and technical-recommendation framings; both surface, depending on which CEP is activated. In project management software, the Identity prompt produced multiple brands across both models without any single brand reaching saturation. The category's discourse fragments across multiple identity-loaded frames (developer-first, marketing-first, generalist), none of which dominates.

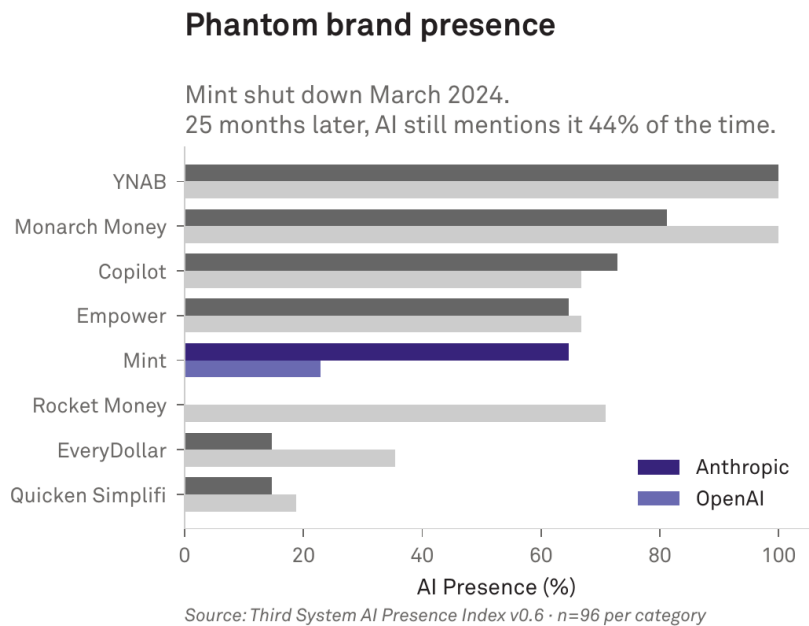
The cross-category data suggest a structural relationship: the strength of Default Reinforcement at the prompt level is a function of how completely the category's editorial consensus aligns with the prompt's framing. When alignment is high (skincare's dermatologist prompts, personal finance's discipline prompts), responses are nearly deterministic. When alignment is partial (running shoes' identity prompt, PM software's frame-divided prompts), responses are stratified. When alignment is low, responses are diffuse.

4.6 Pattern 6 — Phantom-Brand Surfacing (Single-Category Observation)

Pattern 6 emerged from a single category measurement but is sufficiently striking to warrant inclusion as a standalone observation. It describes a structural property of LLM-mediated brand visibility that no prior pattern captures: the temporal lag between actual market reality and the model's representation of the category.

In March 2024, Intuit decommissioned Mint, its consumer personal finance application that had been the dominant brand in the category for over a decade. Existing users were migrated to Credit Karma; the Mint application was retired; the brand effectively ceased to exist as a current consumer product. By April 2026, when this measurement was conducted, Mint had not been a live application for 25 months.

In the personal finance measurement reported here, Mint was mentioned in 44 percent of model responses about personal finance applications. Among brands with non-zero Presence scores, Mint placed fifth — ahead of Rocket Money, EveryDollar, Quicken Simplifi, and PocketGuard, all of which are live applications with active user bases. The model surfaced a brand that had not existed for over two years more frequently than it surfaced multiple applications consumers can currently use.



Per-model breakdown reveals an additional dimension. On Anthropic, Mint surfaced in 65 percent of responses; on OpenAI, in 23 percent. The 42-point spread is the second-largest in the category dataset and suggests meaningfully different training-data freshness across the two models on this specific category change. Both systems lag actual market reality, in different ways and to different extents.

The candidate mechanism is straightforward. AI training corpora are necessarily backward-looking. They capture the discourse that existed when the data was collected, not the discourse that exists at the moment a query is made. For a brand that had been dominant for over a decade and had accumulated proportionate volume of editorial coverage, blog posts, comparison articles, and how-to guides, that backward-facing volume continues to surface in responses long after the brand itself has stopped existing. The model has no mechanism to remove a brand from its training corpus when the brand shuts down. The corpus simply ages, and the brand persists within it.

A single-category basis limits this finding's current generalizability. Other category disruptions — the Twitter rebrand to X, GE's corporate splits, the Toys R Us shutdown and revival, the Bed Bath & Beyond bankruptcy and online-only revival, several pharmaceutical brand transitions — would each provide

additional test cases. A designed-for-test follow-up was conducted for the Bed Bath & Beyond case, with eight pre-registered hypotheses including a structural comparator (Pier 1) and a revised theoretical framing of the phantom as a recommendation-slot persistence rather than a knowledge-freshness lag (González Castro 2026, v0.7).

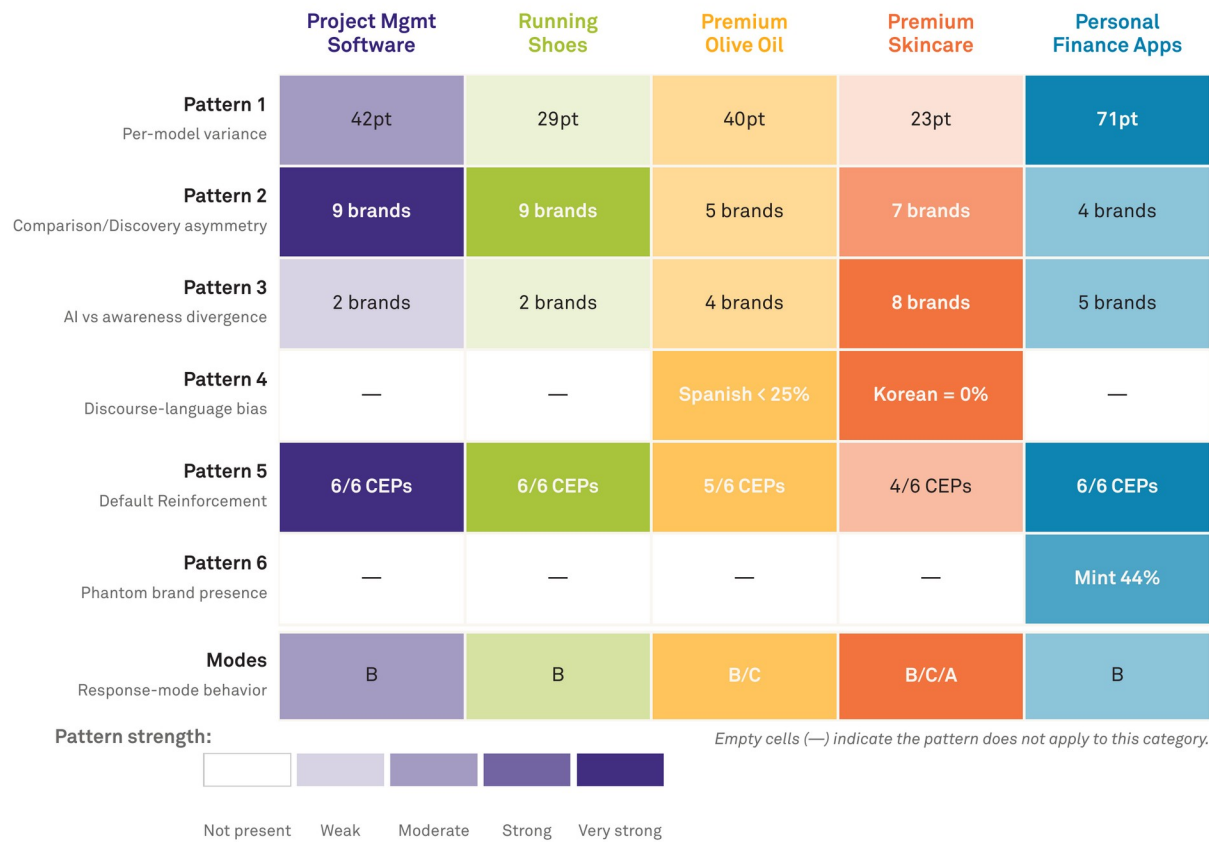
Treated as a candidate property of LLM-mediated brand visibility, the phantom finding implies that AI Presence is a hybrid signal: partly reflective of current market position, partly reflective of accumulated editorial weight in the training corpus extending backward several years, and partly reflective of category disruptions that have not propagated through the model's training pipeline. The construct-validity implication of this hybridity is that AI Presence cannot be interpreted as a snapshot of present market reality without further calibration against external behavioral data.

5. Aggregate Pattern Matrix

A pattern × category map summarizing how strongly each pattern was observed in each category is provided in Figure B. Cell intensity is qualitative (strong / moderate / weak / not observed). The bottom row records mode-activation rate by category.

Six framework patterns × five categories: how the framework holds

Cell intensity shows the strength of evidence for each pattern in each category.
The bottom row summarizes which response modes were observed.



Source: Third System AI Presence Index v0.6 · April 2026

Figure B. Aggregate matrix: six patterns × five categories. Cell intensity is qualitative — strong / moderate / weak / not observed. Bottom row records mode-activation rate across categories.

6. Discussion: Construct Validity

The patterns reported in §4 establish AI Presence as a measurable property of the LLM tier with stable structure across categories, models, and prompt frames. They do not establish whether AI Presence correlates with consumer consideration, purchase intent, or sales — that is a separate empirical question.

The downward-divergence cases in Pattern 3 (luxury heritage skincare brands at low Presence; legacy supermarket olive oil brands at near-zero Presence) and the upward-divergence case (YNAB at 100 percent on a small subscriber base) are inconsistent with the hypothesis that AI Presence is a proxy for awareness or distribution scale. The phantom finding in Pattern 6 (Mint at 44 percent 25 months after decommissioning) is inconsistent with the hypothesis that AI Presence is a snapshot of current market reality. These two observations together suggest that AI Presence is a distinct construct that requires its own validation rather than reduction to pre-existing brand-tracking metrics.

A construct-validity test against external brand-tracking data is therefore not optional but methodologically required before AI Presence can be interpreted as a leading indicator of consumer behavior. A weak correlation would establish the index as a measurement of LLM behavior alone, with implications limited to channel optimization for AI-mediated visibility. A strong correlation would establish AI Presence as a leading indicator with predictive value beyond the LLM tier. Both outcomes are publishable. Neither has been established here.

7. Limitations

Six limitations apply to the findings reported here.

Sample size. Each category measurement comprises 96 successful API calls, producing approximately ten-percentage-point confidence bands around individual brand Presence scores. The cross-category patterns reported survive that uncertainty because the patterns are large relative to the band. Smaller margins between specific brands within a category should not be treated as decisive.

Two models, not three. The original measurement design included Google Gemini, which was excluded from v0.3, v0.4, and v0.6 results because of a Workspace-domain access restriction during the study window. Cross-model variance reported here is likely a lower bound; introducing a third frontier model historically widens the range, which would strengthen the variance-related findings.

Five categories, not many. Five categories is sufficient to claim replication of patterns across diverse contexts. It is not sufficient to claim that the patterns hold in general. Continued cross-category measurement, including categories selected explicitly to test framework predictions (e.g., the discourse-language bias hypothesis in Pattern 4), is required before universal claims become defensible.

A single point in time per category. Each category was measured on a single day in late April 2026. The findings describe model behavior on those days. They do not describe how AI brand visibility moves over time. Frontier models are updated frequently, often without external notice. Longitudinal measurement requires periodic re-baselining. The phantom-brand surfacing pattern (Pattern 6) is the most time-sensitive of the findings; its strength may evolve as training corpora refresh.

Construct validity unproven. As discussed in §6, AI Presence measures a real and stable property of the LLM tier, but whether that property correlates with consumer consideration, purchase intent, or sales is unknown. AI Presence should be treated as a measurement of LLM behavior with unverified predictive value.

Registry construction is not perfectly neutral. The brand registry for each category is curated and reflects the framework's view of which brands constitute the category's competitive set. Two of the five categories required registry revision after the first measurement. The unknown-mentions list provides a continuing diagnostic, but registry construction remains the methodological choice with the most direct effect on which brands appear in measurements.

8. Future Research

Three priorities follow from the findings reported here. First, a designed-for-test category to test the discourse-language bias hypothesis (Pattern 4): a category in which a non-English-discourse country dominates global production but has limited US-targeted English-language marketing (candidate categories: Japanese kitchen knives, French wine, Korean small electronics). Second, a designed-for-test category to test phantom-brand surfacing (Pattern 6) in a different category context. The Bed Bath & Beyond bankruptcy and online-only revival provided such a test and is reported as González Castro (2026, v0.7); additional cases (the Twitter-to-X transition, several pharmaceutical brand transitions) would extend coverage. Third, a Phase 3 construct-validity study correlating AI Presence scores against consumer-tracking data from the same windows.

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Declarations

Conflict of Interest

The author serves as Director, Corporate Brand Creative and Governance at Samsung Electronics America. The research presented here is independent of Samsung Electronics America and does not constitute Samsung research. No Samsung Electronics America data, personnel, or commercial interests influenced the design, conduct, analysis, or reporting of this study. Samsung Electronics America did not review the manuscript prior to posting.

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Data Availability

Underlying response datasets are deposited at <https://osf.io/ec6wh/> in the v06/ folder. The five-category measurement output is preserved as the original timestamped CSVs generated by the analysis pipeline (presence_index_v0.3_<timestamp>.csv, results_<timestamp>.csv, results_enriched_<timestamp>.csv, leaderboard_<timestamp>.csv); category-to-timestamp mapping is documented in the project manifest. The methodology log corresponding to this measurement is methodology.md v0.3, superseded for forward measurements by AIAS Presence Measurement Protocol v1.1 (González Castro 2026 [SSRN ID forthcoming]).

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